

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

Executive Summary

Total Issues Detected: 115
Critical: 16 | Warning: 99 | Info: 0

[LINQ003] Missing Projection (.Select)

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 183)

```
Assert.Equal("Wanderer Black Hiking Boots", result.Data.ToList().FirstOrDefault().Name)
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
Assert.Equal("Wanderer Black Hiking Boots", result.Data.ToList().FirstOrDefault().Name)
```
```

? Recommendation

Review manual refactoring guidelines.

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 183)

```
Assert.Equal("Wanderer Black Hiking Boots", result.Data.ToList().FirstOrDefault().Name)
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
Assert.Equal("Wanderer Black Hiking Boots", result.Data.ToList().FirstOrDefault().Name)
```
```

? Recommendation

Review manual refactoring guidelines.

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

[EF004] Synchronous DB Call in Async Method

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 183)

```
result.Data.ToList().FirstOrDefault()
```

AI Optimization & Guidance:

AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

Performance Impact

Refactored Solution

```
```csharp
result.Data.ToList().FirstOrDefault()
```
```

Recommendation

Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 211)

```
Assert.Contains("Alpine", result.Data.ToList().FirstOrDefault().Name)
```

AI Optimization & Guidance:

AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

Performance Impact

Refactored Solution

```
```csharp
Assert.Contains("Alpine", result.Data.ToList().FirstOrDefault().Name)
```
```

Recommendation

Review manual refactoring guidelines.

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 211)

```
Assert.Contains("Alpine", result.Data.ToList().FirstOrDefault().Name)
```

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
Assert.Contains("Alpine", result.Data.ToList().FirstOrDefault().Name)  
```

? Recommendation
Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 211)

```
result.Data.ToList().FirstOrDefault()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
result.Data.ToList().FirstOrDefault()  
```

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 284)

```
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogTypeId)
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

? Performance Impact

? Refactored Solution

```
```csharp
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogTypeId)
```
```

? Recommendation

Review manual refactoring guidelines.

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 284)

```
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogTypeId)
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogTypeId)
```
```

? Recommendation

Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 284)

```
result.Data.ToList().FirstOrDefault()
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

```
result.Data.ToList().FirstOrDefault()
...
```

### ? Recommendation  
Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 285)

```
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogBrandId)
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp  
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogBrandId)  
...
```

? Recommendation
Review manual refactoring guidelines.

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 285)

```
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogBrandId)
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp  
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogBrandId)
...
```

### ? Recommendation  
Review manual refactoring guidelines.

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

## [EF004] Synchronous DB Call in Async Method

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 285)

```
result.Data.ToList().FirstOrDefault()
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
result.Data.ToList().FirstOrDefault()
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 313)

```
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogBrandId)
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogBrandId)
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 313)

```
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogBrandId)
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
Assert.Equal(3, result.Data.ToList().FirstOrDefault().CatalogBrandId)
```

### ? Recommendation  
Review manual refactoring guidelines.

### [EF004] Synchronous DB Call in Async Method

Warning

Location: tests/Catalog.FunctionalTests/CatalogApiTests.cs (Line 313)

```
result.Data.ToList().FirstOrDefault()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
result.Data.ToList().FirstOrDefault()
```

### ? Recommendation  
Review manual refactoring guidelines.

### [EF004] Synchronous DB Call in Async Method

Warning

Location: tests/Ordering.FunctionalTests/OrderingApiTests.cs (Line 202)

```
payload.Items.Count()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Performance Impact

### ? Refactored Solution

```
```csharp
payload.Items.Count()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Ordering.Infrastructure/MediatorExtension.cs (Line 11)

```
domainEntities
 .SelectMany(x => x.Entity.DomainEvents)
 .ToList()
```

**AI Optimization & Guidance:**

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
domainEntities
    .SelectMany(x => x.Entity.DomainEvents)
    .ToList()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Ordering.Infrastructure/MediatorExtension.cs (Line 11)

```
domainEntities
 .SelectMany(x => x.Entity.DomainEvents)
 .ToList()
```

**AI Optimization & Guidance:**

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden



# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### Performance Impact

### Refactored Solution

```
```csharp
domainEntities
    .SelectMany(x => x.Entity.DomainEvents)
    .ToList()
```
```

### Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Ordering.Infrastructure/MediatorExtension.cs (Line 11)

```
domainEntities
 .SelectMany(x => x.Entity.DomainEvents)
 .ToList()
```

### AI Optimization & Guidance:

#### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### Performance Impact

### Refactored Solution

```
```csharp
domainEntities
    .SelectMany(x => x.Entity.DomainEvents)
    .ToList()
```
```

### Recommendation

Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Ordering.Infrastructure/MediatorExtension.cs (Line 15)

```
domainEntities.ToList()
 .ForEach(entity => entity.Entity.ClearDomainEvents())
```

### AI Optimization & Guidance:

#### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Performance Impact

### ? Refactored Solution

```
```csharp
domainEntities.ToList()
    .ForEach(entity => entity.Entity.ClearDomainEvents())
...

```

? Recommendation

Review manual refactoring guidelines.

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Ordering.Infrastructure/MediatorExtension.cs (Line 15)

```
domainEntities.ToList()
    .ForEach(entity => entity.Entity.ClearDomainEvents())

```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
domainEntities.ToList()
 .ForEach(entity => entity.Entity.ClearDomainEvents())
...

```

### ? Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Ordering.Infrastructure/MediatorExtension.cs (Line 15)

```
domainEntities.ToList()

```

AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Refactored Solution  
```csharp  
domainEntities.ToList()
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF002] N+1 Query Pattern in Foreach Loop

Critical

Location: src/eShop.ServiceDefaults/OpenApiOptionsExtensions.cs (Line 74)

```
foreach (var link in policy.Links.Where(l => l.Type == "text/html"))
{
 if (!rendered...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
foreach (var link in policy.Links.Where(l => l.Type == "text/html"))
{
 ...
 if (!rendered...

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/eShop.ServiceDefaults/OpenApiOptionsExtensions.cs (Line 110)

```
options.AddOperationTransformer((operation, context, cancellationTokens) =>  
{  
    var metadata = context....
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

? Refactored Solution
```csharp  
options.AddOperationTransformer((operation, context, cancellationToken) =>  
 {  
 var metadata = context....  
 }  
...  
  
### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/eShop.ServiceDefaults/OpenApiOptionsExtensions.cs (Line 110)

```
options.AddOperationTransformer((operation, context, cancellationToken) =>
 {
 var metadata = context....
 }
...
}
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden  
  
### ? Performance Impact  
  
### ? Refactored Solution  
```csharp  
options.AddOperationTransformer((operation, context, cancellationToken) =>
 {
 var metadata = context....
 }
...

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/eShop.ServiceDefaults/OpenApiOptionsExtensions.cs (Line 129)

```
scopes.ToList()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

? Performance Impact

? Refactored Solution

```
```csharp
scopes.ToList()
```
```

? Recommendation

Review manual refactoring guidelines.

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/eShop.ServiceDefaults/OpenApiOptionsExtensions.cs (Line 129)

```
scopes.ToList()
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
scopes.ToList()
```
```

? Recommendation

Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: tests/ClientApp.UnitTests/Services/BasketServiceTests.cs (Line 11)

```
result.Count()
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

```
result.Count()
...
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: tests/ClientApp.UnitTests/Services/OrdersServiceTests.cs (Line 30)

```
result.Count()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp  
result.Count()  
...
```

? Recommendation
Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: tests/ClientApp.UnitTests/Services/CatalogServiceTests.cs (Line 12)

```
catalog.Count()
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp  
catalog.Count()
...
```

### ? Recommendation  
Review manual refactoring guidelines.

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

## [EF004] Synchronous DB Call in Async Method

Warning

Location: tests/ClientApp.UnitTests/Services/CatalogServiceTests.cs (Line 21)

```
catalogBrand.Count()
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
catalogBrand.Count()
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: tests/ClientApp.UnitTests/Services/CatalogServiceTests.cs (Line 30)

```
catalogType.Count()
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
catalogType.Count()
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Services/CatalogAI.cs (Line 41)

```
embeddings.Select(m => new Vector(m.Vector[0..EmbeddingDimensions])).ToList()
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
embeddings.Select(m => new Vector(m.Vector[0..EmbeddingDimensions])).ToList()
```

### ? Recommendation  
Review manual refactoring guidelines.

### [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Catalog.API/Services/CatalogAI.cs (Line 41)

```
embeddings.Select(m => new Vector(m.Vector[0..EmbeddingDimensions])).ToList()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
embeddings.Select(m => new Vector(m.Vector[0..EmbeddingDimensions])).ToList()
```

### ? Recommendation  
Review manual refactoring guidelines.

### [EF005] Client-Side Evaluation Trap

Critical

Location: src/Catalog.API/Infrastructure/CatalogContextSeed.cs (Line 30)

```
context.CatalogBrands.AddRangeAsync(sourceItems.Select(x => x.Brand).Distinct()
 .Where(brandName => bran...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden



# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Performance Impact

### ? Refactored Solution

```
```csharp
context.CatalogBrands.AddRangeAsync(sourceltems.Select(x => x.Brand).Distinct()
    .Where(brandName => bran...
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Catalog.API/Infrastructure/CatalogContextSeed.cs (Line 33)

```
context.CatalogBrands.Count()
```

AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
context.CatalogBrands.Count()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF005] Client-Side Evaluation Trap

Critical

Location: src/Catalog.API/Infrastructure/CatalogContextSeed.cs (Line 36)

```
context.CatalogTypes.AddRangeAsync(sourceltems.Select(x => x.Type).Distinct()
 .Where(typeName => typeNam...
```

AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Refactored Solution  
```csharp  
context.CatalogTypes.AddRangeAsync(sourceItems.Select(x => x.Type).Distinct()
 .Where(typeName => typeNam...
...

? Recommendation
Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Catalog.API/Infrastructure/CatalogContextSeed.cs (Line 39)

```
context.CatalogTypes.Count()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport
content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
context.CatalogTypes.Count()  
...  
  
### ? Recommendation  
Review manual refactoring guidelines.

## [EF005] Client-Side Evaluation Trap

Critical

Location: src/Catalog.API/Infrastructure/CatalogContextSeed.cs (Line 46)

```
sourceItems
 .Where(source => source.Name != null && source.Brand != null && source.Type != null)
...
...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport  
content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

```
```csharp
sourceItems
    .Where(source => source.Name != null && source.Brand != null && source.Type != null)
    ...

### ? Recommendation
Review manual refactoring guidelines.
```

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Catalog.API/Infrastructure/CatalogContextSeed.cs (Line 72)

```
context.CatalogItems.Count()
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
context.CatalogItems.Count()
```
```

? Recommendation

Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 77)

```
api.MapGet("/catalogtypes",
    [ProducesResponseType<ProblemDetails>(StatusCodes.Status400BadRequest, "applicat...
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
api.MapGet("/catalogtypes",
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

... [ProducesResponseType<ProblemDetails>(StatusCodes.Status400BadRequest, "applicat...

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 77)

```
api.MapGet("/catalogtypes",
 [ProducesResponseType<ProblemDetails>(StatusCodes.Status400BadRequest, "applicat...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
api.MapGet("/catalogtypes",
 [ProducesResponseType<ProblemDetails>(StatusCodes.Status400BadRequest, "applicat...
...

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 79)

```
context.CatalogTypes.OrderBy(x => x.Type).ToListAsync()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
context.CatalogTypes.OrderBy(x => x.Type).ToListAsync()  
...

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 79)

```
context.CatalogTypes.OrderBy(x => x.Type).ToListAsync()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
context.CatalogTypes.OrderBy(x => x.Type).ToListAsync()
```

### ? Recommendation  
Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 84)

```
api.MapGet("/catalogbrands",
[ProducesResponseType<ProblemDetails>(StatusCodes.Status400BadRequest, "applica...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
api.MapGet("/catalogbrands",
[ProducesResponseType<ProblemDetails>(StatusCodes.Status400BadRequest, "applica...
```

### ? Recommendation  
Review manual refactoring guidelines.

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 84)

```
api.MapGet("/catalogbrands",
 [ProducesResponseType<ProblemDetails>(StatusCodes.Status400BadRequest, "applica...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
api.MapGet("/catalogbrands",
 [ProducesResponseType<ProblemDetails>(StatusCodes.Status400BadRequest, "applica...
...

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 86)

```
context.CatalogBrands.OrderBy(x => x.Brand).ToListAsync()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
context.CatalogBrands.OrderBy(x => x.Brand).ToListAsync()  
...

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 86)

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

```
context.CatalogBrands.OrderBy(x => x.Brand).ToListAsync()
```

### AI Optimization & Guidance:

#### ### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### ? Performance Impact

#### ### ? Refactored Solution

```
```csharp
context.CatalogBrands.OrderBy(x => x.Brand).ToListAsync()
```
```

#### ### ? Recommendation

Review manual refactoring guidelines.

## [EF005] Client-Side Evaluation Trap

Critical

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 138)

```
root.Where(c => c.Name.StartsWith(name))
```

### AI Optimization & Guidance:

#### ### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### ? Performance Impact

#### ### ? Refactored Solution

```
```csharp
root.Where(c => c.Name.StartsWith(name))
```
```

#### ### ? Recommendation

Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 152)

```
root
 .OrderBy(c => c.Name)
 .Skip(pageSize * pageIndex)
 .Take(pageSize)

```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
root
    .OrderBy(c => c.Name)
    .Skip(pageSize * pageIndex)
    .Take(pageSize)
    ....
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 152)

```
root
 .OrderBy(c => c.Name)
 .Skip(pageSize * pageIndex)
 .Take(pageSize)

```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
root
    .OrderBy(c => c.Name)
    .Skip(pageSize * pageIndex)
    .Take(pageSize)
    ....
```
```

#### ### Recommendation

Review manual refactoring guidelines.



# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 166)

```
services.Context.CatalogItems.Where(item => ids.Contains(item.Id)).ToListAsync()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
services.Context.CatalogItems.Where(item => ids.Contains(item.Id)).ToListAsync()
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 166)

```
services.Context.CatalogItems.Where(item => ids.Contains(item.Id)).ToListAsync()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
services.Context.CatalogItems.Where(item => ids.Contains(item.Id)).ToListAsync()
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF005] Client-Side Evaluation Trap

Critical

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 166)

```
services.Context.CatalogItems.Where(item => ids.Contains(item.Id))
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
services.Context.CatalogItems.Where(item => ids.Contains(item.Id))
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 266)

```
services.Context.CatalogItems
 .Where(c => c.Embedding != null)
 .Select(c => new { Item =...
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
services.Context.CatalogItems
    .Where(c => c.Embedding != null)
    .Select(c => new { Item =...
...
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF005] Client-Side Evaluation Trap

Critical

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 266)

```
services.Context.CatalogItems
 .Where(c => c.Embedding != null)
 .Select(c => new { Item =...
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

## AI Optimization & Guidance:

### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### Performance Impact

### Refactored Solution

```
```csharp
services.Context.CatalogItems
    .Where(c => c.Embedding != null)
    .Select(c => new { Item =...
```
```

### Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 276)

```
itemsWithDistance.Select(i => i.Item).ToList()
```

## AI Optimization & Guidance:

### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### Performance Impact

### Refactored Solution

```
```csharp
itemsWithDistance.Select(i => i.Item).ToList()
```
```

### Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 276)

```
itemsWithDistance.Select(i => i.Item).ToList()
```

## AI Optimization & Guidance:

### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
itemsWithDistance.Select(i => i.Item).ToList()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 280)

```
services.Context.CatalogItems
 .Where(c => c.Embedding != null)
 .OrderBy(c => c.Embedding...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport  
content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
services.Context.CatalogItems
    .Where(c => c.Embedding != null)
    .OrderBy(c => c.Embedding...
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 280)

```
services.Context.CatalogItems
 .Where(c => c.Embedding != null)
 .OrderBy(c => c.Embedding...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
services.Context.CatalogItems
    .Where(c => c.Embedding != null)
    .OrderBy(c => c.Embedding...)
...

```

? Recommendation

Review manual refactoring guidelines.

[EF005] Client-Side Evaluation Trap

Critical

Location: src/Catalog.API/Apis/CatalogApi.cs (Line 280)

```
services.Context.CatalogItems
    .Where(c => c.Embedding != null)
    .OrderBy(c => c.Embedding...)

```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport
content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
services.Context.CatalogItems
 .Where(c => c.Embedding != null)
 .OrderBy(c => c.Embedding...)
...

```

### ? Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/ClientApp/ViewModels/CheckoutViewModel.cs (Line 98)

```
orders.Count()
```

AI Optimization & Guidance:

### ? AI Root Cause Analysis

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
orders.Count()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF005] Client-Side Evaluation Trap

Critical

Location: src/ClientApp/Validations/ValidatableObject.cs (Line 38)

```
.Where(v => !v.Check(Value))
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
.Where(v => !v.Check(Value))
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/WebApp/Services/BasketState.cs (Line 35)

```
(await FetchBasketItemsAsync()).Select(i => new BasketQuantity(i.ProductId, i.Quantity)).ToList()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Refactored Solution  
```csharp  
(await FetchBasketItemsAsync()).Select(i => new BasketQuantity(i.ProductId, i.Quantity)).ToList()
```  
  
### ? Recommendation  
Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/WebApp/Services/BasketState.cs (Line 35)

```
(await FetchBasketItemsAsync()).Select(i => new BasketQuantity(i.ProductId, i.Quantity)).ToList()
```

**AI Optimization & Guidance:**  
### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden  
  
### ? Performance Impact  
  
### ? Refactored Solution  
```csharp  
(await FetchBasketItemsAsync()).Select(i => new BasketQuantity(i.ProductId, i.Quantity)).ToList()
```  
  
### ? Recommendation  
Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/WebApp/Services/BasketState.cs (Line 60)

```
(await FetchBasketItemsAsync()).ToList()
```

**AI Optimization & Guidance:**  
### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden  
  
### ? Performance Impact  
  
### ? Refactored Solution  
```csharp  
(await FetchBasketItemsAsync()).ToList()
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/WebApp/Services/BasketState.cs (Line 60)

```
(await FetchBasketItemsAsync()).ToList()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
(await FetchBasketItemsAsync()).ToList()
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/WebApp/Services/BasketState.cs (Line 60)

```
(await FetchBasketItemsAsync()).ToList()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
(await FetchBasketItemsAsync()).ToList()
```

### ? Recommendation  
Review manual refactoring guidelines.



# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/WebApp/Services/BasketState.cs (Line 73)

```
basketService.UpdateBasketAsync(existingItems.Select(i => new BasketQuantity(i.ProductId, i.Quantity)).ToList())
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
basketService.UpdateBasketAsync(existingItems.Select(i => new BasketQuantity(i.ProductId, i.Quantity)).ToList())
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/WebApp/Services/BasketState.cs (Line 73)

```
existingItems.Select(i => new BasketQuantity(i.ProductId, i.Quantity)).ToList()
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
existingItems.Select(i => new BasketQuantity(i.ProductId, i.Quantity)).ToList()
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Webhooks.API/IntegrationEvents/OrderStatusChangedToPaidIntegrationEventHandler.cs (Line 12)

```
subscriptions.Count()
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

## AI Optimization & Guidance:

### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### Performance Impact

### Refactored Solution

```
```csharp
subscriptions.Count()
```
```

### Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Webhooks.API/IntegrationEvents/OrderStatusChangedToShippedIntegrationEventHandler.cs (Line 12)

```
subscriptions.Count()
```

## AI Optimization & Guidance:

### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### Performance Impact

### Refactored Solution

```
```csharp
subscriptions.Count()
```
```

### Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Webhooks.API/Services/GrantUrlTesterService.cs (Line 22)

```
tokenValues.FirstOrDefault()
```

## AI Optimization & Guidance:

### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Performance Impact

### ? Refactored Solution

```
```csharp
tokenValues.FirstOrDefault()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Webhooks.API/Services/WebhooksRetriever.cs (Line 7)

```
db.Subscriptions.Where(s => s.Type == type).ToListAsync()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
db.Subscriptions.Where(s => s.Type == type).ToListAsync()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Webhooks.API/Services/WebhooksRetriever.cs (Line 7)

```
db.Subscriptions.Where(s => s.Type == type).ToListAsync()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
```

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

```
db.Subscriptions.Where(s => s.Type == type).ToListAsync()
...
```

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/Webhooks.API/Apis/WebHooksApi.cs (Line 13)

```
api.MapGet("/", async (WebhooksContext context, ClaimsPrincipal user) =>
{
    var userId = user.GetUser...
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
api.MapGet("/", async (WebhooksContext context, ClaimsPrincipal user) =>  
{  
 var userId = user.GetUser...  
...  
}

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Webhooks.API/Apis/WebHooksApi.cs (Line 13)

```
api.MapGet("/", async (WebhooksContext context, ClaimsPrincipal user) =>
{
 var userId = user.GetUser...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

```
api.MapGet("/", async (WebhooksContext context, ClaimsPrincipal user) =>
{
    var userId = user.GetUser...
...

```

? Recommendation
Review manual refactoring guidelines.

[EF005] Client-Side Evaluation Trap

Critical

Location: src/Webhooks.API/Apis/WebHooksApi.cs (Line 13)

```
api.MapGet("/", async (WebhooksContext context, ClaimsPrincipal user) =>
{
    var userId = user.GetUser...

```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
api.MapGet("/", async (WebhooksContext context, ClaimsPrincipal user) =>  
{  
 var userId = user.GetUser...  
...

### ? Recommendation  
Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Webhooks.API/Apis/WebHooksApi.cs (Line 16)

```
context.Subscriptions.Where(s => s.UserId == userId).ToListAsync()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

```
```csharp
context.Subscriptions.Where(s => s.UserId == userId).ToListAsync()
```
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Webhooks.API/Apis/WebHooksApi.cs (Line 16)

```
context.Subscriptions.Where(s => s.UserId == userId).ToListAsync()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp
context.Subscriptions.Where(s => s.UserId == userId).ToListAsync()
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF005] Client-Side Evaluation Trap

Critical

Location: src/IntegrationEventLogEF/Services/IntegrationEventLogService.cs (Line 13)

```
Assembly.Load(Assembly.GetEntryAssembly().FullName)
 .GetTypes()
 .Where(t => t.Name.EndsWith(name...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp
Assembly.Load(Assembly.GetEntryAssembly().FullName)
 .GetTypes()

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

```
... .Where(t => t.Name.EndsWith(name...
```

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/IntegrationEventLogEF/Services/IntegrationEventLogService.cs (Line 21)

```
_context.Set<IntegrationEventLogEntry>()  
    .Where(e => e.TransactionId == transactionId && e.State == EventSta...
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport
content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
\_context.Set<IntegrationEventLogEntry>()  
 .Where(e => e.TransactionId == transactionId && e.State == EventSta...  
...

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/IntegrationEventLogEF/Services/IntegrationEventLogService.cs (Line 21)

```
_context.Set<IntegrationEventLogEntry>()
 .Where(e => e.TransactionId == transactionId && e.State == EventSta...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport  
content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
_context.Set<IntegrationEventLogEntry>()
 .Where(e => e.TransactionId == transactionId && e.State == EventSta...

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

...

? Recommendation
Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Identity.API/Services/ProfileService.cs (Line 16)

```
subject.Claims.Where(x => x.Type == "sub").FirstOrDefault()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
subject.Claims.Where(x => x.Type == "sub").FirstOrDefault()  
```

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/Identity.API/Services/ProfileService.cs (Line 23)

```
claims.ToList()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
claims.ToList()  
```

? Recommendation
Review manual refactoring guidelines.

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Identity.API/Services/ProfileService.cs (Line 23)

```
claims.ToList()
```

AI Optimization & Guidance:

AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

Performance Impact

Refactored Solution

```
```csharp
claims.ToList()
```
```

Recommendation

Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Identity.API/Services/ProfileService.cs (Line 23)

```
claims.ToList()
```

AI Optimization & Guidance:

AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

Performance Impact

Refactored Solution

```
```csharp
claims.ToList()
```
```

Recommendation

Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Identity.API/Services/ProfileService.cs (Line 30)

```
subject.Claims.Where(x => x.Type == "sub").FirstOrDefault()
```

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

AI Optimization & Guidance:

AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

Performance Impact

Refactored Solution
```csharp  
subject.Claims.Where(x => x.Type == "sub").FirstOrDefault()  
```

Recommendation
Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Identity.API/Services/ProfileService.cs (Line 39)

```
subject.Claims.Where(c => c.Type == "security_stamp").Select(c => c.Value).SingleOrDefault()
```

AI Optimization & Guidance:

AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

Performance Impact

Refactored Solution
```csharp  
subject.Claims.Where(c => c.Type == "security\_stamp").Select(c => c.Value).SingleOrDefault()  
```

Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: tests/Ordering.UnitTests/Domain/SeedWork/ValueObjectTests.cs (Line 148)

```
c.ToList()
```

AI Optimization & Guidance:

AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

? Performance Impact

? Refactored Solution

```
```csharp
c.ToList()
```
```

? Recommendation

Review manual refactoring guidelines.

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: tests/Ordering.UnitTests/Domain/SeedWork/ValueObjectTests.cs (Line 148)

```
c.ToList()
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
c.ToList()
```
```

? Recommendation

Review manual refactoring guidelines.

[EF002] N+1 Query Pattern in Foreach Loop

Critical

Location: src/Catalog.API/IntegrationEvents/EventHandling/
OrderStatusChangedToAwaitingValidationIntegrationEventHandler.cs (Line 15)

```
foreach (var orderStockItem in @event.OrderStockItems)
{
    var catalogItem = catalogContext.CatalogIte...
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

? Refactored Solution

```
```csharp
foreach (var orderStockItem in @event.OrderStockItems)
{
 var catalogItem = catalogContext.CatalogIte...
}
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF002] N+1 Query Pattern in Foreach Loop

Critical

Location: src/Catalog.API/IntegrationEvents/EventHandling/OrderStatusChangedToPaidIntegrationEventHandler.cs (Line 13)

```
foreach (var orderStockItem in @event.OrderStockItems)
{
 var catalogItem = catalogContext.CatalogIte...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
foreach (var orderStockItem in @event.OrderStockItems)
{
    var catalogItem = catalogContext.CatalogIte...
}
```

? Recommendation

Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/Ordering.API/Application/Behaviors/ValidatorBehavior.cs (Line 23)

```
validationResults
    .SelectMany(result => result.Errors)
    .Where(error => error != null)
    ...
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

⚠ Performance Impact

⚠ Refactored Solution

```
```csharp
validationResults
 .SelectMany(result => result.Errors)
 .Where(error => error != null)
...

```

### ⚠ Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Ordering.API/Application/Behaviors/ValidatorBehavior.cs (Line 23)

```
validationResults
 .SelectMany(result => result.Errors)
 .Where(error => error != null)
...

```

### AI Optimization & Guidance:

#### ⚠ AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport  
content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ⚠ Performance Impact

### ⚠ Refactored Solution

```
```csharp
validationResults
    .SelectMany(result => result.Errors)
    .Where(error => error != null)
...

```

⚠ Recommendation

Review manual refactoring guidelines.

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Ordering.API/Application/Behaviors/ValidatorBehavior.cs (Line 23)

```
validationResults
    .SelectMany(result => result.Errors)

```

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

```
.Where(error => error != null)
...
```

AI Optimization & Guidance:

AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

Performance Impact

Refactored Solution

```
```csharp
validationResults
 .SelectMany(result => result.Errors)
 .Where(error => error != null)
...

```

### Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Ordering.API/Application/Queries/OrderQueries.cs (Line 27)

```
order.OrderItems.Select(oi => new Orderitem
{
 ProductName = oi.ProductName,
 ...

```

## AI Optimization & Guidance:

### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### Performance Impact

### Refactored Solution

```
```csharp
order.OrderItems.Select(oi => new Orderitem
{
    ProductName = oi.ProductName,
    ...

```

Recommendation

Review manual refactoring guidelines.

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Ordering.API/Application/Queries/OrderQueries.cs (Line 27)

```
order.OrderItems.Select(oi => new Orderitem
{
    ProductName = oi.ProductName,
    ...
})
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
order.OrderItems.Select(oi => new Orderitem
{
 ProductName = oi.ProductName,
 ...
})
```

#### ### ? Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Ordering.API/Application/Queries/OrderQueries.cs (Line 39)

```
context.Orders
 .Where(o => o.Buyer.IdentityGuid == userId)
 .Select(o => new OrderSummary
 {
 ...
 })
```

### AI Optimization & Guidance:

#### ### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### ? Performance Impact

#### ### ? Refactored Solution

```
```csharp
context.Orders
    .Where(o => o.Buyer.IdentityGuid == userId)
    .Select(o => new OrderSummary
    {
        ...
    })
```

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

? Recommendation
Review manual refactoring guidelines.

[EF005] Client-Side Evaluation Trap

Critical

Location: src/Ordering.API/Application/Queries/OrderQueries.cs (Line 39)

```
context.Orders
    .Where(o => o.Buyer.IdentityGuid == userId)
    .Select(o => new OrderSummary
    ...
```

AI Optimization & Guidance:

? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution

```
```csharp
context.Orders
 .Where(o => o.Buyer.IdentityGuid == userId)
 .Select(o => new OrderSummary
 ...
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Ordering.API/Application/Queries/OrderQueries.cs (Line 52)

```
context.CardTypes.Select(c=> new CardType { Id = c.Id, Name = c.Name }).ToListAsync()
```

### AI Optimization & Guidance:

#### ### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### ? Performance Impact

#### ### ? Refactored Solution

```
```csharp
context.CardTypes.Select(c=> new CardType { Id = c.Id, Name = c.Name }).ToListAsync()
...`
```


LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

? Recommendation
Review manual refactoring guidelines.

[LINQ003] Missing Projection (.Select)

Warning

Location: src/Ordering.API/Application/Commands/CreateOrderCommand.cs (Line 70)

```
basketItems.ToOrderItemsDTO().ToList()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
basketItems.ToOrderItemsDTO().ToList()  
```

? Recommendation
Review manual refactoring guidelines.

[EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Ordering.API/Application/Commands/CreateOrderCommand.cs (Line 70)

```
basketItems.ToOrderItemsDTO().ToList()
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
basketItems.ToOrderItemsDTO().ToList()  
```

? Recommendation
Review manual refactoring guidelines.

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

[EF005] Client-Side Evaluation Trap

Critical

Location: src/Ordering.Domain/AggregatesModel/OrderAggregate/Order.cs (Line 161)

```
OrderItems
    .Where(c => orderStockRejectedItems.Contains(c.ProductId))
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
OrderItems  
 .Where(c => orderStockRejectedItems.Contains(c.ProductId))  
...

### ? Recommendation  
Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Identity.API/Quickstart/Account/AccountController.cs (Line 231)

```
schemes
 .Where(x => x.DisplayName != null)
 .Select(x => new ExternalProvider
 ...
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
schemes
 .Where(x => x.DisplayName != null)
 .Select(x => new ExternalProvider
 ...

? Recommendation
Review manual refactoring guidelines.

LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>
Branch: main | Generated: 2026-08-10 14:41 UTC

[EF004] Synchronous DB Call in Async Method

Warning

Location: src/Identity.API/Quickstart/Account/AccountController.cs (Line 231)

```
schemes
    .Where(x => x.DisplayName != null)
    .Select(x => new ExternalProvider
    ...
```

AI Optimization & Guidance:

? AI Root Cause Analysis
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

? Performance Impact

? Refactored Solution
```csharp  
schemes  
 .Where(x => x.DisplayName != null)  
 .Select(x => new ExternalProvider  
 ...

### ? Recommendation  
Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Identity.API/Quickstart/Account/AccountController.cs (Line 249)

```
providers.Where(provider => client.IdentityProviderRestrictions.Contains(provider.AuthenticationScheme)).ToList()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
providers.Where(provider => client.IdentityProviderRestrictions.Contains(provider.AuthenticationScheme)).ToList()
```

### ? Recommendation  
Review manual refactoring guidelines.

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Identity.API/Quickstart/Account/AccountController.cs (Line 249)

```
providers.Where(provider => client.IdentityProviderRestrictions.Contains(provider.AuthenticationScheme)).ToList()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
providers.Where(provider => client.IdentityProviderRestrictions.Contains(provider.AuthenticationScheme)).ToList()
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Identity.API/Quickstart/Account/AccountController.cs (Line 249)

```
providers.Where(provider => client.IdentityProviderRestrictions.Contains(provider.AuthenticationScheme)).ToList()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis  
AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution  
```csharp  
providers.Where(provider => client.IdentityProviderRestrictions.Contains(provider.AuthenticationScheme)).ToList()
```

### ? Recommendation  
Review manual refactoring guidelines.

## [EF005] Client-Side Evaluation Trap

Critical

Location: src/Identity.API/Quickstart/Account/AccountController.cs (Line 249)

```
providers.Where(provider => client.IdentityProviderRestrictions.Contains(provider.AuthenticationScheme))
```

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
providers.Where(provider => client.IdentityProviderRestrictions.Contains(provider.AuthenticationScheme))
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [LINQ003] Missing Projection (.Select)

Warning

Location: src/Identity.API/Quickstart/Account/ExternalController.cs (Line 148)

```
externalUser.Claims.ToList()
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

#### ### Performance Impact

#### ### Refactored Solution

```
```csharp
externalUser.Claims.ToList()
```
```

#### ### Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Identity.API/Quickstart/Account/ExternalController.cs (Line 148)

```
externalUser.Claims.ToList()
```

### AI Optimization & Guidance:

#### ### AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Performance Impact

### ? Refactored Solution

```
```csharp
externalUser.Claims.ToList()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Identity.API/Quickstart/Account/ExternalController.cs (Line 148)

```
externalUser.Claims.ToList()
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
externalUser.Claims.ToList()
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF001] Missing AsNoTracking() in Read-Only Query

Warning

Location: src/Ordering.API/Application/IntegrationEvents/EventHandling/OrderStockRejectedIntegrationEventHandler.cs (Line 10)

```
@event.OrderStockItems
.FindAll(c => !c.HasStock)
.Select(c => c.ProductId)
.ToList(...)
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

# LINQ & EF Core Performance Report

Repository: <https://github.com/dotnet/eShop.git>  
Branch: main | Generated: 2026-08-10 14:41 UTC

### ? Refactored Solution

```
```csharp
@event.OrderStockItems
    .FindAll(c => !c.HasStock)
    .Select(c => c.ProductId)
    .ToList(...)
```
```

### ? Recommendation

Review manual refactoring guidelines.

## [EF004] Synchronous DB Call in Async Method

Warning

Location: src/Ordering.API/Application/IntegrationEvents/EventHandling/  
OrderStockRejectedIntegrationEventHandler.cs (Line 10)

```
@event.OrderStockItems
 .FindAll(c => !c.HasStock)
 .Select(c => c.ProductId)
 .ToList(...)
```

### AI Optimization & Guidance:

### ? AI Root Cause Analysis

AI Evaluation Error: HTTP 403: <!doctype html><meta charset="utf-8"><meta name=viewport content="width=device-width, initial-scale=1"><title>403</title>403 Forbidden

### ? Performance Impact

### ? Refactored Solution

```
```csharp
@event.OrderStockItems
    .FindAll(c => !c.HasStock)
    .Select(c => c.ProductId)
    .ToList(...)
```
```

### ? Recommendation

Review manual refactoring guidelines.